Machine Learning-Based Pre-Diagnosis Tools in Emergency Departments: Predicting Hospitalization, Mortality and Triage Acuity

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***Abstract*—With demographic growth, aging populations, and the recent advent of the Covid-19 pandemic, emergency department staff are increasingly overwhelmed, leading to a decline in the quality of patient care and an alarming rise in hospital mortality rates. Thus, there is a growing need for accurate and reliable pre-diagnostic tools to optimize patient flow and improve the quality of care. While many artificial intelligence-based tools have been tested for pre-diagnosis in emergency services, they are still not effective enough. Moreover, exploring the use of additional indicators to enhance preliminary diagnosis should also be considered. This paper aims to predict the Emergency Severity Index—a standard reference for triaging patients—along with other indicators that clinicians can use to refine preliminary diagnoses, particularly regarding patient hospitalization and mortality. To achieve this, we utilized the MIMIC-IV (Medical Information Mart for Intensive Care) and MIMIC-IV ED (Emergency Department) datasets, applying several machine learning techniques. We propose a new pipeline for these prediction tasks, employing Polynomial Features for feature extraction and Principal Component Analysis for dimensionality reduction. Our proposed methods achieved high accuracy rates, with hospitalization prediction reaching 97.00%, mortality prediction at 86.41%, and triage acuity prediction scoring of 99.80% using Logistic Regression, Random Forest, and Decision Tree classifiers.**

***Index Terms*—Emergency Severity Index, Machine Learning, Mortality, Hospital Admission, MIMIC-IV**

I. Introduction

Emergency Departments (**ED**) worldwide have recently experienced a significant increase in overcrowding [1], presenting unprecedented challenges to healthcare systems. The increasing number of patients seeking urgent medical care, coupled with the need to adhere to infection control protocols due to emerging viruses and maintain physical distancing measures, has added another level of complexity to an already overburdened **ED** system, resulting in longer waiting times for evaluation and treatment, and higher risks of critical situations. Machine Learning (**ML**) has emerged as an innovative and very helpful technology with significant potential in the healthcare sector [2], [3], such as in automatic disease diagnosis, medical imaging, and, more recently, in clinical trial and research. This progress could not have been achieved without the introduction of Electronic Health Records (**EHR**) [4], which are digital versions of a patient's medical history, with lab results, vital signs, medications, and demographic data.

Researchers have provided many solutions to improve emergency care by predicting risks of critical situations such as transfers to the Intensive Care Unit (**ICU**), **ICU** mortality, in-hospital mortality, **ED** re-attendance, and estimating the patients' length of stay in the hospital. The aim of this study is to propose approaches that enable clinical staff to better manage patients in an emergency context. To this end, we predict three outcomes: *mortality*, *hospitalization* and *triage acuity*. Predictions of mortality and hospital admission provide a warning system for nurses, helping them identify patients with higher risks of critical situations to make better and faster decisions. Triage acuity refers to the categorization of patients based on the seriousness of their conditions and the urgency of the required medical care. The Emergency Severity Index (**ESI**), the most widely used triage scale in the United States, categorizes patients into five levels, with level $\mathbf{1}$ being the most urgent and level $\mathbf{5}$ being the least urgent.

In this paper, we propose new approaches for predicting *mortality*, *hospital admission, and discharge*, and develop an *e-triage tool* based on the **ESI** using the Medical Information Mart for Intensive Care IV (**MIMIC-IV**) [5] and the **MIMIC-IV ED** [6] datasets. These datasets provide valuable insights about patients' medical records outside and within of emergency departments, respectively. Our approaches leverage **ML** techniques for data pre-processing, feature extraction with Polynomial Features (**PF**)—a novel method for generating new features based on the non-linear relationships of our variables, Principal Component Analysis (**PCA**) for dimensionality reduction, and classification using Random Forest (**RF**), Logistic Regression (**LR**), and Decision Tree (**DT**) classifiers. We focus on integrating feature extraction techniques into our pipeline, as they have enabled us to obtain better results for predicting all three outcomes.

The article is structured as follows: Section II reviews recent literature on predicting mortality, hospitalization, and triage acuity based on the **ESI**. Section III introduces the proposed

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approach for each outcome. The evaluation protocol is detailed in Section IV, along with a description of the datasets used. The obtained results are presented in Section V. Finally, the conclusion in Section VI provides an opportunity to highlight the conceptual and methodological implications of our study and suggests avenues for future research.

II. Background

A. Fundamentals

The design of a system aimed at predicting mortality, hospital admission, and triage acuity typically adheres to a general workflow composed of the following steps:

*1) **Data Extraction and Cleaning**:* This initial stage consists in defining the research problem and identifying the cohort of interest. Thus, a subset of the original dataset is extracted, including pertinent attributes such as demographic details, medical diagnoses, laboratory results, and hospital stay history.

*2) **Data Pre-processing**:* This phase involves various procedures to tailor and refine the data for **ML** applications. For **EHR**s, the steps are contingent upon the data category. Standard processes include the imputation of missing values using diverse strategies, outlier detection and resolution, categorical variable encoding, and normalization.

*3) **Feature Extraction**:* This involves employing multiple methods to convert variables into more significant features. Although not commonly included in similar workflows, we incorporated it in ours for a richer data representation.

*4) **Dimensionality Reduction**:* This can be characterized as the process of reducing the variable count while retaining the most critical information. Various techniques may be employed, such as Principal Component Analysis (**PCA**) and Linear Discriminant Analysis (**LDA**).

*5) **Classification**:* While regression might be applied for mortality risk and general deterioration prediction, classification remains the principal method for such analyses. Different **ML** algorithms may be utilized, such as Logistic Regression (**LR**), Random Forest (**RF**), Gradient Boosting (**GB**), and Support Vector Machine (**SVM**) [7].

LR performs well when the feature-to-target relationship is nearly linear. It is also fast to train and straightforward. **RF** is more effective with high-dimensional data. Its ensemble averaging mechanism makes it robust against overfitting and efficient at managing noisy data. **SVM** is especially advantageous due to its kernel trick, which facilitates the conversion of data into a higher-dimensional space for non-linear separation. **GB** excels with complex, non-linear data and manages various feature types, offering an ensemble strategy to mitigate overfitting. Additionally, neural networks [8], such as Multi-Layer Perceptrons (**MLP**s) are also used. While they are efficient in identifying complex patterns, they necessitate significant data and computational power and are often intricate to interpret.

Previous research has delved into numerous **ML** algorithms for mortality, hospitalization, and triage acuity prediction. Our methodology introduces an innovative feature extraction step with polynomial features to detect complex non-linear interdependencies among inputs, thus yielding a more thorough data depiction. We also employ **PCA** for dimensionality reduction to improve our models' interpretability and streamline the underlying structure while conserving vital information. This not only augments the efficiency of the model but also clarifies the decision-making process for healthcare professionals.

B. Related Work

*1) **Hospitalization Prediction**:* entails assessing the probability of patients requiring hospital admission for continuous care or further treatment.

Xie et al. [9] proposed to compare the performances of several prediction models for hospitalization, critical care and **ED** re-attendance in the context of emergency departments using **ML** algorithms such as **LR**, **RF**, **GB** and **MLP**. For the performance evaluation, they used **MIMIC-IV** and **MIMIC-IV ED** datasets. To our knowledge, this is the only work that discussed hospitalization prediction in **ED**s using **MIMIC-IV**. However, there are numerous studies on this topic using other datasets.

Hong et al. [10] utilized **ML** methods to predict hospital admissions in emergency department triage, employing nine binary classifiers including **LR**, eXtreme Gradient Boosting (**XGBoost**) and Deep Neural Networks (**DNN**).

Mowbray et al. [11] predicted emergency disposition for elderly patients by training classification algorithms such as **LR**, **RF**, **SVM**, Classification And Regression Tree (**CART**) and Gradient Boosted Trees (**GBT**).

Ahmed et al. [12] suggested employing three **ML** approaches: **XGBoost**, **Adaboost**, and **MLP**, each integrated with Tabu Search, for their hospitalization prediction.

While this topic has been extensively studied, there is a scarcity of studies that have utilized the **MIMIC-IV** dataset. However, the pipeline proposed by Xie et al. [9] provides a suitable framework to ease the data extraction from the original dataset. Based on their benchmark, we conducted our study to contribute to the existing literature in this field.

*2) **Mortality Prediction**:* consists in evaluating the risk of patients facing critical complications or not surviving during their hospital stay. Our study uses the **MIMIC-IV** and **MIMIC-IV ED** datasets for this purpose, and we subsequently review existing methods that have also employed these datasets for mortality prediction.

Gupta et al. [13] presented a data processing pipeline for predicting numerous clinical prediction tasks using both **ICU** and non-**ICU** data. They trained multiple **ML** models (such as **LR**, **RF**, **GB**, **XGBoost**) and *deep learning* models (such as Recurrent Neural Networks (**RNN**) with Long Short Term Memory (**LSTM**) cells).

Shu et al. [14] predicted **ICU** stay, 3-days-death, 7-days-death, and 30-days-death after emergency admission for Ischemic heart disease using scoring models established by the AutoScore and using **ML** algorithms for classification.

Hu et al. [15] used multiple **ML** models using **LR**, **DT**, **RF**, *k*-Nearest Neighbors (*k*-**NN**), **ADAB**, Naive Bayes, **LDA**,

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NN and **XGBoost** to develop a system to predict in-hospital mortality in septic patients re-admitted to the **ICU**.

Deng et al. [16] used time-series *deep learning* models such as **RNN**, Gated Recurrent Unit (**GRU**) and **LSTM** with the combination of three widely used scoring systems (**APACHE II**, Sequential Organ Failure Assessment (**SOFA**) and Simplified Acute Physiology Score **SAPS II**) to predict in-hospital mortality, prolonged length of stay, and 30-day re-admission.

3) ***Triage Acuity Prediction****:* usually, nurses are responsible for performing triage in emergency departments and undergo training sessions to effectively use different triage scales such as **ESI**, Manchester Triage System (**MTS**) and Canadian Triage and Acuity Scale (**CTAS**). As **ESI** is the triage scale used in our dataset, we present only the studies adopting this scaling method for triage acuity.

Mutegeki et al. [17] developed a **ML** model for predicting patient triage acuity, employing a range of classifiers including **DT**, **RF**, Histogram-Based Gradient Boosting (**HGBT**), **XGBoost**, and Extreme Learning Machine (**ELM**).

Ivanov et al. [18] developed an automated approach using **XGBoost** and Clinical Natural Language Processing (**C-NLP**) techniques.

Ahmed et al. [19] utilized three metaheuristic optimization algorithms – Simulated Annealing (**SA**), Adaptive Simulated Annealing (**ASA**), and Grid Search (**GS**) – to fine-tune the parameters of two **ML** models, **XGBoost** and Categorical Boosting (**CaB**), for predicting triage acuity. This approach led to the development of six distinct methods.

Moreover, Soufi et al. [20] proposed a hybrid approach that combines rule-based reasoning (**RBR**) and fuzzy logic (**FLC**).

III. Proposed Approaches

This study encompasses three primary classification tasks, depicted in Fig. 1. It outlines the architecture developed for predicting patient mortality, hospitalization, and triage acuity, with each task involving a series of fundamental steps.

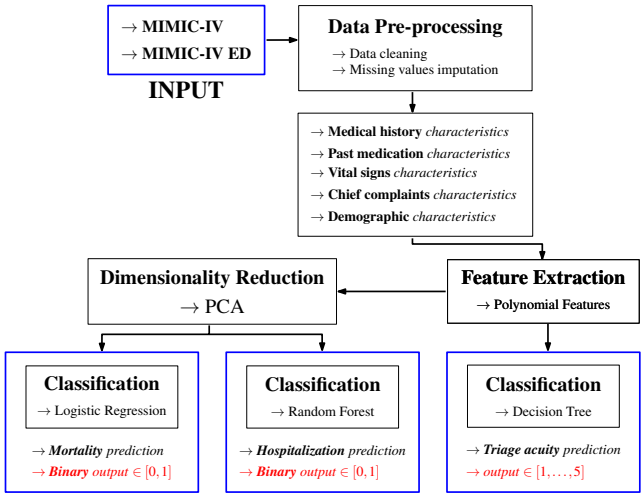


Fig. 1. Overview of the proposed approaches.

A. Data Extraction

The data was extracted using the pipeline proposed by Xie et al. [9] where they used both **MIMIC-IV** and **MIMIC-IV ED** datasets. The features chosen were extracted based on the recommendation of healthcare professionals and informaticians who possess a deep understanding of the unprocessed data and the operational aspects of the emergency department. Based on the extracted subsets, we chose to predict three outcomes: *hospitalization*, *mortality*, and *triage acuity*. For each prediction task, we opted for not applying oversampling or undersampling methods; hence, three subsets were extracted to obtain balanced datasets. They were trained using the same attributes, except for mortality prediction where the hospitalization status was added. All information about each outcome is shown in Table I.

TABLE I
DETAILS ABOUT EACH EXTRACTED SUBSET.

Subset for prediction	$\sum$ *Patients*	$\sum$ *Samples*	$\sum$ *Variables*
Hospital admission	123000	200000	93
Mortality	9520	9786	93
Triage acuity	5575	5875	95

B. Data Pre-processing

1) ***Data Cleaning****:* We excluded variables with a percentage of missing values greater than 30%. We also removed variables indicating dates, such as admission and discharge.

2) ***Missing Values Imputation****:* For classification using **EHR** data, it is essential to meticulously address missing values, outliers, and inaccurate records. Outliers in vital signs and lab tests, identified as values outside the clinically-established reasonable physiological range, were treated as missing. We utilized thresholds from Harutyunyan et al.'s source code [21], supported by clinical expertise, for outlier detection. To impute these missing values, the Multivariate Imputation by Chained Equations (**MICE**) [22] method was applied. This iterative method generates multiple imputed datasets, thus enhancing the imputation process over Simple Imputation techniques that can lead to underestimation of variability and introduction of bias. **MICE** is especially able to impute continuous variables such as vital signs due to its ability to preserve their statistical properties. For categorical variables, including chief complaints and demographic information like age, we utilized mode imputation. This ensures the plausibility of the imputed values and maintains the integrity of the data distribution.

C. Feature Extraction

In predictive modeling, capturing the intricate non-linear interactions between input features is crucial, often necessitating transformations like polynomial feature generation. This process enriches the feature space, aiming to uncover the underlying complexities within the data. For example, a set of features $\{x_1, x_2, x_3, \ldots\}$, when transformed with polynomial terms of degree $d = 2$, expands to $\{x_1^2, x_2^2, x_3^2, x_1x_2, x_1x_3, x_2x_3, \ldots\}$. However, adding polynomial features for better accuracy can

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make models less interpretable, a serious concern in healthcare where clear understanding is essential for clinicians and patients.

D. Dimensionality Reduction

PCA, a cornerstone technique for dimensionality reduction, was originally formulated by Karl Pearson. It serves to project high-dimensional data onto a lower-dimensional subspace, preserving as much variance as possible. This is operationalized through principal components, which capture the predominant variance within the data. The transformation is mathematically defined by Eq. 1:

$$Z = W^T X \tag{1}$$

Where Z refers to the matrix of transformed data, W to the matrix of *eigenvectors* and X to the matrix of original data. Our application of **PCA** aims to enhance computational efficiency by reducing the dimensions from $n = 4465$ to $m = 20$ for hospital admission prediction, and from $n = 4656$ to $m = 83$ for mortality prediction. **PCA** enhances the efficiency and visual interpretation of intricate **EHR** data for pattern discovery in predictive modeling, yet the interpretation of the resulting principal components, as aggregates of original data features, poses a subtle challenge.

E. Classification

The classification task in our study involves multiple *supervised* **ML** algorithms (refer to Fig. 1). We approach hospitalization prediction as a binary classification, assigning a 0 for *discharge* and 1 for *admission*. Similarly, mortality prediction is binary, with 0 indicating *survival* and 1 indicating *death*. The triage acuity prediction is treated as a *multi-class* classification task, predicting levels from 1 (most urgent) to 5 (least urgent). The most effective models were **RF**, **LR**, and **DT** for each task respectively.

1) ***Random Forest (RF):*** employs an ensemble of decision trees, each one is constructed from a randomly sampled subset of the training data. The ensemble's prediction is the mode of the individual trees' predictions, thus enhancing the robustness and accuracy of the classification.

2) ***Logistic Regression (LR):*** serves as a probabilistic binary classifier. It estimates the probability of an observation belonging to a default class (labeled 1), predicting this class if the probability exceeds 50% and the alternative class (labeled 0) otherwise.

3) ***Decision Tree (DT):*** forms the building blocks of the **RF** model. It structures the data in a tree-like model of decisions, using a series of binary choices to infer the classification, making it intuitive and relatively easy to interpret. The decision tree's construction relies on information gain as a criterion.

IV. Experimentation And Evaluation

A. Data Description

Our study used the **MIMIC-IV** and **MIMIC-IV ED** datasets, encompassing de-identified **EHR**s from Beth Israel Deaconess Medical Center, Boston, from 2008 to 2019. **MIMIC-IV** includes hospital and **ICU** admissions data, while **MIMIC-IV ED** focuses on **ED** visits. Study variables, including comorbidities measured by the Charlson Comorbidity Index (**CCI**) and Elixhauser Comorbidity Index (**ECI**), are detailed in Table II. Additionally, Table III provides insights into selected **ED** data features such as mean age, average **ED** stay, and triage acuity levels, and their distribution in relation to hospitalization and mortality outcomes.

TABLE II
CATEGORIES OF THE USED VARIABLES.

Information category	*Variables*
Patient History	→ Past **ED** visits. → Past hospitalizations. → Past **ICU** admissions. → Comorbidity.
Related to the triage	→ Demographics (*age* and *gender*). → Vital signs measured at triage. → Triage-chief complaints.
Before the **ED** disposition	→ Last measurements of the vitals signs. → **ED** length of stay. → Counts of medication reconciliation. → Counts of medication prescription.

TABLE III
OVERVIEW OF KEY FEATURES INFLUENCING HOSPITALIZATION AND MORTALITY PREDICTIONS.

		Hospitalization		**Mortality**	
		Discharge	Admission	Survival	Death
Age		46.29	60	52.62	72.31
Triage acuity	1	2251	9328	247	1528
	2	19530	47660	1591	2011
	3	64685	39758	2619	552
	4	12239	691	349	0
	5	496	24	11	0
Gender	Female	57565	50560	2636	2287
	Male	42435	49440	2257	2606
ED Length of stay (hrs)		0.3	9.6	4.79	5.74

B. Validation Protocol

The input data was partitioned into 67% for training and 33% for testing. For mortality and hospitalization predictions, the Area Under the Curve (**AUC**) metric was used, assessing the model's ability to distinguish between positive and negative cases, with values near 100% indicating higher accuracy. Triage acuity prediction relied on accuracy for performance evaluation. The study also analyzed feature importance to understand each variable's contribution in the **ML** models.

V. Results And Discussion

The evaluation of our models is presented in Tables IV, V, and VI. For hospital admission predictions, our model attained an accuracy of 97.00%, as detailed in Table VII. In mortality predictions, the application of **PCA** enhanced model

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TABLE IV
COMPARISON FOR HOSPITAL ADMISSION PREDICTION IN TERMS OF **AUC** WITH *state-of-the-art* METHODS.

Method	*Dataset*	*Details*	**AUC**
Mowbray et al. [11]	Eight **ED** sites in Canada	GBT	80.00%
Ahmed et al. [12]	**ED** visits in the US	T-ADAB	95.40%
Xie et al. [9]	**MIMIC-IV** & **MIMIC-IV ED**	RF	81.90%
Hong et al. [10]	Yale New Haven Health system	XGBoost	91.00%
Proposed method	**MIMIC-IV & MIMIC-IV ED**	**PF, PCA, RF**	**97.00%**

TABLE V
COMPARISON FOR MORTALITY PREDICTION IN TERMS OF **AUC** WITH *state-of-the-art* METHODS.

Method	*Dataset*	*Details*	**AUC**
Gupta et al. [13]	MIMIC-IV	Hybrid LSTM	86.00%
Shu et al. [14]		AutoScore	79.00%
Hu et al. [15]		RF	81.00%
Deng et al. [16]		GRU	87.00%
Proposed method	**MIMIC-IV & MIMIC-IV ED**	**PF + PCA + LR**	**86.00%**

TABLE VI
COMPARISON FOR TRIAGE ACUITY PREDICTION IN TERMS OF ACCURACY WITH *state-of-the-art* METHODS.

Method	*Dataset*	*Details*	*Accuracy*
Mutegeki et al. [17]	Yale New Haven Health system	HGBT	70.00%
Ivanov et al. [18]	Two participating hospitals	KATE triage model	78.50%
Soufi et al. [20]	**ED** of Imam Reza Hospital in Iran	RBR + FLC	99.40%
Ahmed et al. [19]	**ED** visits in the US	ASA-CaB	83.30%
Proposed method	**MIMIC-IV & MIMIC-IV ED**	**PF + DT**	**99.80%**

efficiency by reducing the number of necessary components. This method outperformed most standard approaches and is further elaborated in comparison with Deng et al. [16]. The combination of **PCA** and **PF** significantly improved the **LR** performance, as confirmed by the confusion matrices in Tables VII and VIII.

For triage acuity predictions, **PF** was used, yielding high accuracy as demonstrated in Table IX. Our method surpassed those in related works, even with different datasets. Fig. 2 illustrate our classification methods against the **PF** and **LR**, and **PCA** and **LR** pipelines, showcasing our approach's superior accuracy with various **ML** algorithms.

TABLE VII
CONFUSION MATRIX FOR *hospital admission prediction*.

	Admission	**Discharge**
Admission	**98.30**	1.70
Discharge	3.82	**96.18**

TABLE VIII
CONFUSION MATRIX FOR *mortality prediction*.

	Survival	**Death**
Survival	**82.70**	17.30
Death	9.26	**90.74**

The analysis of the most significant factors that affect hospital admission and triage acuity predictions reveals several important common features. These include the length of stay in the **ED**, patient age, the pain scale recorded at triage, and

TABLE IX
CONFUSION MATRIX FOR *triage acuity*.

	1	**2**	**3**	**4**	**5**
1	**100.0**	0.00	0.00	0.00	0.00
2	0.00	**99.74**	0.00	0.26	0.00
3	0.00	0.26	**99.74**	0.00	0.00
4	0.00	0.00	0.00	**100.0**	0.00
5	0.26	0.00	0.00	0.00	**99.74**

the count of medication prescriptions during the current **ED** visit. These factors are critical in reflecting the urgency and complexity of the patient's condition. Vital signs measurement is critical during both triage and the **ED** stay. Initial readings are vital for immediate patient assessment, while continuous monitoring in the **ED** is imperative for ongoing patient care management.

VI. CONCLUDING REMARKS

To conclude, our research significantly advances the fields of healthcare and emergency prediction by focusing on hospitalization and mortality outcomes. It also validates the **ESI**, which is crucial for triage acuity assessment.

We utilized **PF** and **PCA** to enhance predictive performance. Nonetheless, given the complexity of healthcare and the potential challenges with interpretability, coupled with the multifaceted nature of healthcare data, it is essential to strive for greater model transparency. This is part of an ethical and accountable approach, especially when predicting critical outcomes that can drive important decisions such as patient prioritization and resource allocation. Such an approach is

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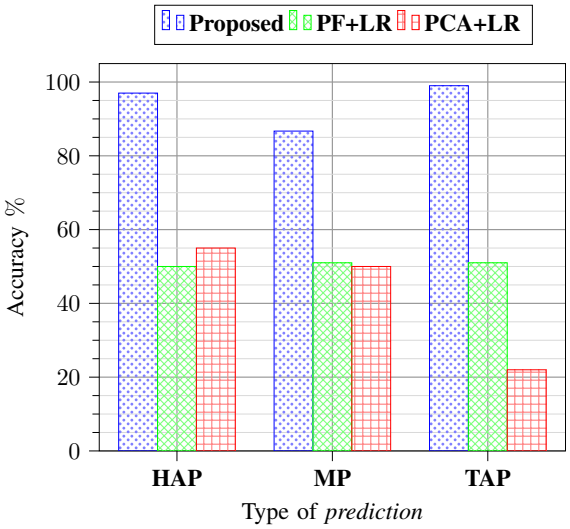


Fig. 2. Comparing three methods for Hospital Admission Prediction (**HAP**), Mortality Prediction (**MP**) and Triage Acuity Prediction (**TAP**).

necessary to adhere to the stringent ethical and regulatory standards within healthcare.

Looking ahead, collaboration with medical experts and the expansion to diverse datasets and healthcare contexts are imperative. Analyzing and presenting significant features across datasets will enrich our understanding of both universal and context-specific determinants in our predictive models. Furthermore, the exploration of predictive indicators based on commonly available data aims to create a more versatile and adaptive predictive system for healthcare applications among different facilities.

Ultimately, we plan to develop an intuitive web-based application that implements various models and will be easy to use by medical staff. We also plan to improve the models' performance using datasets from Canadian hospitals.

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REFERENCES

[1] G. Savioli, I. F. Ceresa, N. Gri, G. Bavestrello Piccini, Y. Longhitano, C. Zanza, A. Piccioni, C. Esposito, G. Ricevuti, and M. A. Bressan, "Emergency department overcrowding: understanding the factors to find corresponding solutions," *Journal of personalized medicine*, vol. 12, no. 2, p. 279, 2022.

[2] M. Javaid, A. Haleem, R. P. Singh, R. Suman, and S. Rab, "Significance of machine learning in healthcare: Features, pillars and applications," *International Journal of Intelligent Networks*, vol. 3, pp. 58–73, 2022.

[3] M. Arbane, M. Belkhelfa, Y. Yaddaden, N. Beder, and S. B. Belhaouari, "Deep learning based method for alzheimer's disease stages classification using mri images," in *2022 2nd International Conference on Advanced Electrical Engineering (ICAEE)*. IEEE, 2022, pp. 1–6.

[4] R. S. Rudin, M. W. Friedberg, P. Shekelle, N. Shah, and D. W. Bates, "Getting value from electronic health records: research needed to improve practice," *Annals of Internal Medicine*, vol. 172, no. 11_Supplement, pp. S130–S136, 2020.

[5] A. Johnson, L. Bulgarelli, T. Pollard, S. Horng, L. A. Celi, and R. Mark, "Mimic-iv," *PhysioNet. Available online at: https://physionet.org/content/mimiciv/1.0/(accessed August 23, 2021)*, 2020.

[6] A. Johnson, L. Bulgarelli, T. Pollard, L. Celi, R. Mark, and S. Horng, "Mimic-iv-ed (version 1.0). physionet," 2021.

[7] A. Naemi, T. Schmidt, M. Mansourvar, M. Naghavi-Behzad, A. Ebrahimi, and U. K. Wiil, "Machine learning techniques for mortality prediction in emergency departments: a systematic review," *BMJ open*, vol. 11, no. 11, p. e052663, 2021.

[8] S. Purushotham, C. Meng, Z. Che, and Y. Liu, "Benchmarking deep learning models on large healthcare datasets," *Journal of biomedical informatics*, vol. 83, pp. 112–134, 2018.

[9] F. Xie, J. Zhou, J. W. Lee, M. Tan, S. Li, L. S. Rajnthern, M. L. Chee, B. Chakraborty, A.-K. I. Wong, A. Dagan *et al.*, "Benchmarking emergency department prediction models with machine learning and public electronic health records," *Scientific Data*, vol. 9, no. 1, p. 658, 2022.

[10] W. S. Hong, A. D. Haimovich, and R. A. Taylor, "Predicting hospital admission at emergency department triage using machine learning," *PloS one*, vol. 13, no. 7, p. e0201016, 2018.

[11] F. Mowbray, M. Zargoush, A. Jones, K. de Wit, and A. Costa, "Predicting hospital admission for older emergency department patients: Insights from machine learning," *International Journal of Medical Informatics*, vol. 140, p. 104163, 2020.

[12] A. Ahmed, O. Ashour, H. Ali, and M. Firouz, "An integrated optimization and machine learning approach to predict the admission status of emergency patients," *Expert Systems with Applications*, vol. 202, p. 117314, 2022.

[13] M. Gupta, B. Gallamoza, N. Cutrona, P. Dhakal, R. Poulain, and R. Beheshti, "An extensive data processing pipeline for mimic-iv," in *Machine Learning for Health*. PMLR, 2022, pp. 311–325.

[14] T. Shu, J. Huang, J. Deng, H. Chen, Y. Zhang, M. Duan, Y. Wang, X. Hu, and X. Liu, "Development and assessment of scoring model for icu stay and mortality prediction after emergency admissions in ischemic heart disease: a retrospective study of mimic-iv databases," *Internal and Emergency Medicine*, vol. 18, no. 2, pp. 487–497, 2023.

[15] C. Hu, L. Li, Y. Li, F. Wang, B. Hu, and Z. Peng, "Explainable machine-learning model for prediction of in-hospital mortality in septic patients requiring intensive care unit readmission," *Infectious Diseases and Therapy*, vol. 11, no. 4, pp. 1695–1713, 2022.

[16] Y. Deng, S. Liu, Z. Wang, Y. Wang, Y. Jiang, and B. Liu, "Explainable time-series deep learning models for the prediction of mortality, prolonged length of stay and 30-day readmission in intensive care patients," *Frontiers in Medicine*, vol. 9, p. 933037, 2022.

[17] H. Mutegeki, A. Nahabwe, J. Nakatumba-Nabende, and G. Marvin, "Interpretable machine learning-based triage for decision support in emergency care," in *2023 7th International Conference on Trends in Electronics and Informatics (ICOEI)*. IEEE, 2023, pp. 983–990.

[18] O. Ivanov, L. Wolf, D. Brecher, E. Lewis, K. Masek, K. Montgomery, Y. Andrieiev, M. McLaughlin, S. Liu, R. Dunne *et al.*, "Improving ed emergency severity index acuity assignment using machine learning and clinical natural language processing," *Journal of Emergency Nursing*, vol. 47, no. 2, pp. 265–278, 2021.

[19] A. Ahmed, M. Al-Maamari, M. Firouz, and D. Delen, "An adaptive simulated annealing-based machine learning approach for developing an e-triage tool for hospital emergency operations," *arXiv preprint arXiv:2212.11892*, 2022.

[20] M. D. Soufi, T. Samad-Soltani, S. S. Vahdati, and P. Rezaei-Hachesu, "Decision support system for triage management: A hybrid approach using rule-based reasoning and fuzzy logic," *International journal of medical informatics*, vol. 114, pp. 35–44, 2018.

[21] H. Harutyunyan, H. Khachatrian, D. C. Kale, G. Ver Steeg, and A. Galstyan, "Multitask learning and benchmarking with clinical time series data," *Scientific data*, vol. 6, no. 1, p. 96, 2019.

[22] I. R. White, P. Royston, and A. M. Wood, "Multiple imputation using chained equations: issues and guidance for practice," *Statistics in medicine*, vol. 30, no. 4, pp. 377–399, 2011.

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